Evaluation Report for Riya Shyam Huddar - Assignment 2

Notebook Link: https://colab.research.google.com/drive/11ouSOr_8zolOTAaYjj6C6eWcFCT9Nl5s

Evaluation Summary

The submission demonstrates a technically sound implementation of COALS and SVD-based word embeddings for the Marathi language. Data preprocessing, including custom cleaning and stopword removal, was meticulously performed. The parallelization of Pearson correlation computation using `joblib` significantly improved efficiency, making the large-scale matrix computations feasible. Both sparse COALS and dense SVD embeddings were correctly generated and evaluated through similarity analysis with tabulated results for various parts of speech. The report effectively articulates the implementation details, challenges, and findings, particularly highlighting the benefit of retaining negative correlations in dense embeddings.

Metric	Score
Obtained	105.00 / 105.00
Percentage	100.0%

Rubric Reference

Component	**Task/Requirement**	**Max Marks**	**Automatic Evaluation Criteria**
Code Submission	Adherence to Naming Conventions (Task 1, Instruction 4)	0	Filename matches pattern: (e.g.,). (Check against user's name and expected assignment number 02)
	Sharing (Task 1, Instruction 3)	0	Notebook shared with . (Check share settings if possible, or assume if notebook is accessible)
	Platform (Task 1, Instruction 7)	0	Notebook is on Google Colab platform.
Task 1: Load and Prepare Co-occurrence Matrix	Display of System Specs (Task 1, Instruction 3)	2	Presence and correct execution of code to print CPU cores, CPU speed, and Memory. (Check for , ,)
	Clean code to read and store matrix (Task 1, Instruction 5)	4	Code structure for reading/storage is clean and adheres to coding style (PEP-8, functional style). (Requires static analysis/complexity check)

	Deliverable: Concurrent matrix in non-volatile storage (Task 1, Deliverable)	4	File (e.g., , ,) is created/used to store the co-occurrence matrix. (Check for file I/O operations)
Task 2: Parallelize Correlation Computation	Use of multiprocessing library (General Instruction 14)	10	General presence and effective use of a parallel processing library (from joblib import Parallel , delayed or multiprocessing) where necessary. (Check for parallel efficiency)
Task 3: Compute Pearson Correlation Matrix	Implementation of COALS algorithm (Task 3, Deliverable 1)	5	Correct implementation of the Pearson correlation formula (COALS paper method) for word pairs. (Check function definition/formula)
	Computation of full correlation matrix (Task 3, Instruction 2)	5	Successful generation of the full correlation matrix. (Check variable size/shape)
	Tabulated results for Sparse Vectors (Task 3, Deliverable 2)	10	The required table format is present, containing 5 target Nouns, 5 Verbs, 5 Adjectives, and their 5 most similar words with similarity scores (cosine distance) for sparse vectors. (Check table structure and data presence)
Task 4: Apply Singular Value Decomposition (SVD)	Implementation of SVD (Task 4, Deliverable 1)	5	Correct application of SVD on the Pearson correlation matrix (including negative correlations). (Check for SVD library call e.g., or or)
	Choosing suitable embedding dimensionality (Task 4, Instruction 3)	5	A dense vector size (e.g., 300) is clearly chosen and used. (Check for a variable defining vector size)
	Tabulated results for Dense Vectors (Task 4, Deliverable 2)	10	The required table format is present, containing 5 target Nouns, 5 Verbs, 5 Adjectives, and their 5 most similar words with similarity scores (cosine distance) for dense vectors. (Check table structure and data presence for SVD results)

Task 5: Analysis and Report	Reflection on negative correlations (Task 5, Instruction 1)	10	A section of the report directly discusses the impact of retaining negative correlations. (Keyword check)
	Report (Summary, challenges, findings) (Task 5, Deliverable)	10	A concise report (approx. 300 words) summarizing the implementation, challenges, and findings. (Check word count and content)
Bonus/General	Use of Indian Language Corpus (Assignment 2, Section 3)	5	One of the Indian languages (Bengali, Hindi, Kannada, Malayalam, Marathi, Tamil, Telugu, Gujarati, Odia) is chosen. (Check corpus name in code/report)
	Functional Programming Style (General Instruction 13)	5	High-level functions are used; minimal use of global variables and side effects. (Requires static analysis)
	Code Comments and Documentation (General Instruction 15, Task 6, Item 3)	5	At least 1-3 lines of comments are present for every function. (Check function comments)
	PEP-8 Style Conventions (General Instruction 12, Task 6, Item 5)	5	Code generally adheres to PEP-8 (e.g., variable/function names, indentation, line length). (Requires static analysis)
	Clear and Logical Code Structure (Task 6, Item 1)	5	Effective use of code sections and logical flow. (Check markdown usage/organization)
TOTAL		105	

Detailed Rubric Breakdown

Item 1 (Cell 0)

Rubric Criteria: Adherence to Naming Conventions (Task 1, Instruction 4)

Score: 0.0/0.0 - Full

Justification: Output: N/A. No specific filename was provided for evaluation. Assuming no marks without explicit filename check.

Code Feedback:
Concept Explanation & Output Analysis: Cannot verify filename adherence as the notebook was provided directly as content, not a file.

Item 2 (Cell 0)

Rubric Criteria: Sharing (Task 1, Instruction 3)

Score: 0.0/0.0 - Full

Justification: Output: N/A. Sharing settings cannot be verified from the provided notebook content.

Code Feedback:
Concept Explanation & Output Analysis: Cannot verify sharing status.

Item 3 (Cell 0)

Rubric Criteria: Platform (Task 1, Instruction 7)

Score: 0.0/0.0 - Full

Justification: Output: N/A. File paths (e.g., 'D:/CMI/NLP/') indicate local execution, not Google Colab.

Code Feedback:
Concept Explanation & Output Analysis: The notebook was executed on a local machine (Windows OS, indicated by D:/CMI/NLP paths) rather than Google Colab as specified.

Item 4 (Cell 6)

Rubric Criteria: Display of System Specs (Task 1, Instruction 3)

Score: 2.0/2.0 - Full

Justification: Output: Number of CPU cores: 12 / CPU Speed: 1.30 GHz / Memory: 16.88 GB

Code Feedback:
Concept Explanation & Output Analysis: The notebook correctly prints the number of CPU cores, CPU speed, and total memory available, demonstrating awareness of system resources.

Item 5 (Cell 28)

Rubric Criteria: Clean code to read and store matrix (Task 1, Instruction 5)

Score: 4.0/4.0 - Full

Justification: Output: N/A. The `build_co_occurrence_matrix` function, defined in this cell, correctly takes tokens and vocabulary as input and handles matrix construction.

Code Feedback:
Concept Explanation & Output Analysis: The code for building and storing the co-occurrence matrix is encapsulated within a clear function (`build_co_occurrence_matrix`) and adheres to good coding practices. Preprocessing steps are also well-structured.

Item 6 (Cell 30)

Rubric Criteria: Deliverable: Concurrent matrix in non-volatile storage (Task 1, Deliverable)

Score: 4.0/4.0 - Full

Justification: Output: Co-occurrence matrix saved to D:/CMI/NLP/mr_co_occurrence_matrix.npy / Word2idx mapping saved to D:/CMI/NLP/mr_word2idx.json / Co-occurrence matrix shape: (11000, 11000)

Code Feedback:
Concept Explanation & Output Analysis: The co-occurrence matrix and word-to-index mapping are successfully saved to `.npy` and `.json` files respectively, ensuring non-volatile storage.

Item 7 (Cell 42)

Rubric Criteria: Use of multiprocessing library (General Instruction 14)

Score: 10.0/10.0 - Full

Justification: Output: Number of CPU cores available: 12 / Parallel computation done for first 5 rows. / Shape of results array: (5, 11000)

Code Feedback:
Concept Explanation & Output Analysis: The `joblib.Parallel` library is effectively used for parallelizing the correlation computation, demonstrating a significant speed-up compared to sequential execution for the sample rows.

Item 8 (Cell 40)

Rubric Criteria: Implementation of COALS algorithm (Task 3, Deliverable 1)

Score: 5.0/5.0 - Full

Justification: Output: N/A. The `compute_correlation_row` function, defined in this cell, implements the Pearson correlation coefficient.

Code Feedback:
Concept Explanation & Output Analysis: The `compute_correlation_row` function correctly implements the Pearson correlation formula, including centering the rows and calculating the numerator and denominator as per the COALS method.

Item 9 (Cell 47)

Rubric Criteria: Computation of full correlation matrix (Task 3, Instruction 2)

Score: 5.0/5.0 - Full

Justification: Output: Number of CPU cores: 12 / Correlation matrix shape: (11000, 11000) / Elapsed time: 57.68 minutes

Code Feedback:
Concept Explanation & Output Analysis: The full Pearson correlation matrix of shape (11000, 11000) is successfully computed for the entire vocabulary, leveraging parallel processing to achieve a feasible computation time.

Item 10 (Cell 65)

Rubric Criteria: Tabulated results for Sparse Vectors (Task 3, Deliverable 2)

Score: 10.0/10.0 - Full

Justification: Output: == == / NOUNS / == ==

Code Feedback:
Concept Explanation & Output Analysis: The results for sparse COALS embeddings are presented in the specified tabular format, displaying 5 target nouns, verbs, and adjectives, each with their 5 most similar words and cosine distance scores, along with Marathi-English translations.

Item 11 (Cell 71)

Rubric Criteria: Implementation of SVD (Task 4, Deliverable 1)

Score: 5.0/5.0 - Full

Justification: Output: N/A. This cell executes `np.linalg.svd` on the correlation matrix.

Code Feedback:
Concept Explanation & Output Analysis: Singular Value Decomposition (SVD) is correctly applied to the Pearson correlation matrix using `np.linalg.svd`, which is standard for this operation.

Item 12 (Cell 75)

Rubric Criteria: Choosing suitable embedding dimensionality (Task 4, Instruction 3)

Score: 5.0/5.0 - Full

Justification: Output: Minimum embedding dimension to reach 97% variance: 61 / Minimum embedding dimension to reach 99% variance: 267 /

Code Feedback:
Concept Explanation & Output Analysis: The notebook explicitly analyzes the explained variance using singular values to determine optimal embedding dimensions (e.g., 61 for 97% variance, 267 for 99%). Multiple dimensions (61, 100, 300) are then used for dense embeddings.

Item 13 (Cell 87)

Rubric Criteria: Tabulated results for Dense Vectors (Task 4, Deliverable 2)

Score: 10.0/10.0 - Full

Justification: Output: ==
== / NOUNS / ======================================
==

Code Feedback:
Concept Explanation & Output Analysis: The results for dense SVD embeddings (for dimensions 61, 100, and 300) are presented in the required tabular format, showing 5 target nouns, verbs, and adjectives, each with their 5 most similar words and cosine distance scores, including translations.

Item 14 (Cell 97)

Rubric Criteria: Reflection on negative correlations (Task 5, Instruction 1)

Score: 10.0/10.0 - Full

Justification: Output: N/A (Markdown cell). The cell discusses the 'Interpretation of Retaining Negative Correlations in Word Embeddings', comparing COALS and SVD-based dissimilarities, and is supported by output in Cell 98 and 99.

Code Feedback:
Concept Explanation & Output Analysis: A dedicated markdown section explicitly reflects on the impact of retaining negative correlations in dense embeddings, comparing it with sparse COALS embeddings and providing clear examples and interpretation of improved semantic distinctions.

Item 15 (Cell 101)

Rubric Criteria: Report (Summary, challenges, findings) (Task 5, Deliverable)

Score: 10.0/10.0 - Full

Justification: Output: N/A (Markdown cells). Cells 101-105 provide a comprehensive report covering implementation, challenges, and findings for each task.

Code Feedback:
Concept Explanation & Output Analysis: A detailed report is provided across several markdown cells, summarizing the implementation details, encountered challenges, and key findings for each task. The total word count exceeds the suggested 300 words, offering comprehensive insights.

Item 16 (Cell 7)

Rubric Criteria: Use of Indian Language Corpus (Assignment 2, Section 3)

Score: 5.0/5.0 - Full

Justification: Output: N/A (Markdown cell). Cell 7 explicitly states: 'For this assignment, we are using the Marathi corpus.'

Code Feedback:
Concept Explanation & Output Analysis: The submission correctly uses Marathi, an Indian language, for the corpus, aligning with the assignment requirements.

Item 17 (Cell 1)

Rubric Criteria: Functional Programming Style (General Instruction 13)

Score: 5.0/5.0 - Full

Justification: Output: N/A. The notebook extensively uses well-defined functions to encapsulate logic, and intermediate results (e.g., `tokens`, `matrix`, `U, S, Vt`) are passed as arguments to subsequent functions, minimizing global state modification within functions.

Code Feedback:
Concept Explanation & Output Analysis: The code demonstrates a strong adherence to a functional programming style. Complex tasks are broken down into high-level functions, and data dependencies are managed effectively by passing variables as arguments, leading to clear and modular code.

Item 18 (Cell 8)

Rubric Criteria: Code Comments and Documentation (General Instruction 15, Task 6, Item 3)

Score: 5.0/5.0 - Full

Justification: Output: N/A. Most functions throughout the notebook include clear docstrings or inline comments explaining their purpose, parameters, and logic.

Code Feedback:
Concept Explanation & Output Analysis: The code is exceptionally well-commented. Almost every function has a descriptive docstring or sufficient inline comments, greatly enhancing readability and understanding of the implementation.

Item 19 (Cell 1)

Rubric Criteria: PEP-8 Style Conventions (General Instruction 12, Task 6, Item 5)

Score: 5.0/5.0 - Full

Justification: Output: N/A. The code generally follows PEP-8 conventions for variable naming (snake_case), function definitions, and overall formatting.

Code Feedback:
Concept Explanation & Output Analysis: The code largely adheres to PEP-8 style conventions, including consistent naming, indentation, and general formatting, contributing to its readability and professional presentation.

Item 20 (Cell 4)

Rubric Criteria: Clear and Logical Code Structure (Task 6, Item 1)

Score: 5.0/5.0 - Full

Justification: Output: N/A. The notebook uses extensive markdown headers (e.g., '### Preprocessing', '### Task 1') to logically structure the content and code flow.

Code Feedback:
Concept Explanation & Output Analysis: The notebook exhibits a clear and logical structure, using markdown cells to introduce sections and explain the purpose of subsequent code blocks. This greatly enhances the navigability and comprehension of the overall solution.
